

JK LIFESTYLE DIGITAL ECOSYSTEM

Technical Architecture, Execution Roadmap & Costing Plan

Website | Super App | Enterprise Platform (ERP) | Clinical | Data, Events & AI

Prepared for	JK Lifestyle Ltd. (Group Management Office)
Prepared as	CTO / Technical Architect board proposal and execution baseline
Baseline	JK Lifestyle Technical Development Plan, September 2026
Planning horizon	24 months
Budget currency	BDT (Bangladeshi Taka)

Budget status: planning baseline, not a vendor quotation. Approval should follow HR salary validation, cloud sizing, license choice and supplier quotations.

1. Executive Summary

Architectural baseline. The program is designed as one shared digital ecosystem rather than separate applications. A shared JK ID, API Gateway and cross-cutting control plane sit above two deliberately separated enterprise cores: Odoo Community as the commercial/financial system of record, and a custom JK ERP/Clinical platform for differentiated clinical, workforce and operational workflows. Odoo Community processes Product, Inventory/Warehouse, Procurement, POS, outlet sales, online/e-commerce sales, customer invoicing and Accounting. The custom ERP/Clinical platform owns Doctor Profile, Patient/Appointment workflow, OPD/EMR, Care Plan, Clinical Billing workflow, ARM, Attendance and Payroll calculation. Financial outcomes from custom workflows are posted to Odoo through controlled APIs/events so there is only one financial ledger. The architecture remains modular/bounded-domain, event-enabled where useful, and preserves clean seams for later service extraction.

Metric	Base Planning Value	Comment
Program duration	24 months	Overlapping phase delivery; two-week sprint cadence
Build-team peak	~14 active technical FTE at peak	Ramp up by phase; no separate PO, Data/AI, ML or moderator headcount
24-month development & specialist cost	BDT 26.9M	Loaded in-house cost plus outsourced Odoo/UIUX support; consolidated roles absorbed by Tech Lead
Cloud / platform run cost	BDT 10.0M	Gradually increases with App, media and AI usage
Other delivery costs	BDT 11.5M	AI/API, SaaS, security, equipment, migration/training, recruitment
Base spend before contingency	BDT 48.4M	Cash plan for approved work
Contingency reserve	BDT 5.8M (12%)	Held centrally; released only through change/risk control
Total planning envelope	BDT 54.3M (~5.43 crore)	Recommended approval ceiling for the 24-month baseline

Important exclusions. Payment/MFS transaction charges are treated as business transaction costs tied to GMV and are not included in the project envelope. The ERP baseline now assumes Odoo Community (open-source), so no Odoo Enterprise license fee is included. Any paid hosting, third-party Odoo add-ons, implementation/support subscription, taxes/VAT, or unusually high video/AI consumption should be budgeted separately if applicable.

2. Planning Assumptions & Guardrails

- Target design scale: 11 outlets initially, 500-700 daily patients, 5,000 peak concurrent users and 50,000+ registered online customers in Year 1. Infrastructure is sized progressively, not pre-purchased for full scale.
- JK Tech is treated as having no pre-existing engineering bench at project start; recruitment is therefore on the critical path.
- Odoo Community is the commercial/financial source of truth for Product, Inventory/Warehouse, Procurement, POS, outlet and online/e-commerce sales, customer invoices and Accounting. Custom JK ERP/Clinical owns Doctor, Appointment, OPD/EMR, Care Plan, Clinical Billing, ARM, Attendance and Payroll workflows; integration is API/event based.
- No big-bang migration. Odoo/Excel/POS/website/clinical migrations use staging rehearsal, parallel run, reconciliation and outlet/module-by-module cut-over.
- Two-week sprints are mandatory. Every sprint ends with a demo and targeted UAT; each major go-live requires regression, performance, security and business acceptance gates.
- Identity, custom ERP/Clinical and DevOps remain in-house. Odoo specialist work may be outsourced with knowledge transfer; Product Owner, Data/AI, ML/AI and Community Moderation remain consolidated under the Tech Lead / Architect in this budget.
- Cloud and third-party budgets are reviewed monthly with alerts at 70%, 85% and 100% of the approved envelope.

3. System Architecture & Technical Design

Architecture objective. Build one coherent digital operating platform with explicit ownership boundaries, a shared identity/control plane, API-first integration and event capture from day one. The design prioritizes business continuity, maintainability, health-data protection and future service extraction without paying day-one microservice operational cost.

3.1 Architecture decisions & guardrails

Decision	Choice	Technical rationale
Architecture style	Modular monolith / bounded domains	5-6 domain-aligned modules/services; extract only when scaling, compliance or workload isolation requires it.
ERP strategy	Hybrid: Odoo Community + Custom ERP/Clinical	Odoo Community owns commercial/financial processing (Product, Inventory/Warehouse, Procurement, POS, outlet and online/e-commerce sales, customer invoices and Accounting). Custom JK ERP/Clinical owns Doctor/Patient, Appointment, OPD/EMR, Care Plan, Clinical Billing workflow, ARM, Attendance and Payroll calculation.
Cloud posture	Hybrid-leaning-cloud	Managed cloud for applications/data, with local resilience for clinical/outlet continuity where internet dependence is unacceptable.
Integration model	API-first + domain events	No direct cross-domain database access; synchronous APIs for commands/queries, events for decoupled downstream reactions.
Identity	One JK ID	A single party identity can hold Customer, Patient, Member, Employee or Vendor roles without duplicate customer records.
Migration	Parallel run, no big bang	Rehearse in staging, shadow/parallel run, reconcile, preserve rollback and cut over outlet/module by module.
Delivery	Continuous UAT + gated releases	Two-week sprints, demos, targeted UAT, security/performance/DR evidence before major go-live.

3.2 Layered system architecture

Layer	Primary components	Responsibility
Experience / Channel	Super Website; Super App; Staff Apps; Custom ERP/Clinical Console; Odoo Community POS/Admin	Customer, workforce, clinical and commercial operations interfaces. Website/App remain JK-owned experiences; Odoo processes commercial transactions behind controlled

Layer	Primary components	Responsibility
		adapters/APIs.
Shared Foundation / Control Plane	JK ID/SSO; API Gateway; RBAC; Consent; Notification; CMS; Search; Payment; Loyalty; Audit	Central identity, policy enforcement, common communications and cross-platform governance.
Business Domains	Commercial/Financial ERP (Odoo); Clinical/Care; HR/ARM/Workforce; Membership/Fitness; Community/Education; Data/AI	Bounded contexts with explicit ownership. Odoo is authoritative for commercial/financial records; custom domains own differentiated operational workflows.
Data, Events & Intelligence	PostgreSQL; Redis; RabbitMQ; Object Storage/CDN; Warehouse; AI/ML services	Operational persistence, cache/session, asynchronous events, media, analytics and model workloads.
Integration	Payment/MFS; SMS/Email/WhatsApp; Health Connect/HealthKit; wearable APIs/BLE; Sinbad; biometric/lab devices	Adapters isolate external providers and Odoo integration contracts from custom domain logic, preserving replaceability and controlled coupling.

- Boundary rule: each domain owns its data; other domains consume APIs or events, never shared tables.
- Evolution rule: Clinical and AI/Food Recognition are the first likely candidates for independent service extraction because of health-data isolation and inference workloads.
- Data rule: events required for alerts, analytics and future prediction are captured from day one; historical data cannot be retrofitted later.
- Migration rule: no production cutover without rehearsal, reconciliation, rollback and business-owner sign-off.

3.3 Domain ownership & ERP boundary

Domain	System of record / ownership	Scope
Identity & Shared Platform	Custom / JK Tech	JK ID, authentication, RBAC, consent, notification, audit, API gateway, integration contracts and identity mapping to Odoo customer/partner records.
Commercial & Financial ERP	Odoo Community + JK Tech integration boundary	Product, Inventory/Warehouse, Procurement, POS, outlet sales, online/e-commerce sales processing, customer invoices, payment/reconciliation and Accounting/ledger.
Clinical & Care Plan	Custom / JK Tech + Clinical SME	Doctor Profile, Patient, Appointment request/process, OPD, EMR, consent, Care Plan authoring, tasks, alerts and clinician workflow.
HR / ARM / Workforce	Custom / JK Tech	ARM, employee/workforce data, Attendance and Payroll calculation/approval workflow. Approved financial

Domain	System of record / ownership	Scope
		summaries post to Odoo Finance.
Digital Commerce Experience & Booking	Custom channels + Odoo transaction engine	Next.js Website/Super App own UX, discovery, cart/checkout orchestration and non-commercial bookings; Odoo processes product sales/orders, stock and financial transaction records.
Membership / Fitness / CRM	Custom / JK Tech	Membership lifecycle, classes, profile and CRM/Customer 360 read model over shared JK ID and cross-system references.
Community / Education / Franchise	Custom / JK Tech	Community feed, moderation console, LMS/video, franchise/distribution and field-operation workflows.
Data / BI / AI	Custom / JK Tech	Event capture, warehouse, dashboards, adherence analysis and future model-serving layer; consumes Odoo and custom-domain events/read models, not direct shared tables.

ERP boundary. Odoo Community is a core commercial/financial application, not a generic shared database. It is the system of record for products/stock, POS and sales transactions, customer financial invoices, payments/reconciliation and the accounting ledger. The custom JK ERP/Clinical platform owns clinical and workforce context. A clinical invoice is created/approved in the custom Clinical workflow, then posted to Odoo for accounting/payment/ledger; approved payroll results follow the same rule. Website and Super App remain JK-owned experiences but route e-commerce transaction processing to Odoo through an adapter/API. No custom service writes directly to Odoo tables; integrations use supported APIs/contracts/events and carry idempotency/correlation identifiers.

3.4 Identity & data architecture

Entity	Purpose	Primary consumers
party (JK ID)	Identity root: name, contact, authentication identity and consent linkage	All channels and custom domains; mapped to Odoo partner/customer reference
party_role	Customer / Patient / Member / Employee / Vendor role linkage	CRM, Clinical, Membership, HR/ARM
odoo_product / lot / stock reference	Odoo-owned product, batch/lot, warehouse and stock movement identifiers exposed through adapter/read models	Website/App catalogue read model, fulfillment, traceability, reporting

Entity	Purpose	Primary consumers
appointment	Patient, doctor, request/status, schedule, location and service context	Custom ERP/Clinical, App/Website booking, notifications
care_plan / target / task	Plan versions, typed targets and daily/weekly tasks	Clinical authoring, App planner, alerts
device_reading	Time-series glucose, weight, BP, steps, sleep, source and timestamp	App, Alert Engine, Warehouse
clinical_bill	Clinical services, professional/service context, approval status and Odoo invoice reference	Custom Clinical workflow; Odoo financial posting/payment
attendance / payroll_run	Attendance inputs, payroll calculation, approval status and Odoo journal/payable reference	Custom HR/ARM/Payroll; Odoo Finance
commercial_order reference	Odoo sales/POS/e-commerce order ID, channel, customer mapping and fulfillment/payment status	Website/App read models, CRM, analytics, support

- Traceability: product_batch links origin → stock ledger movements → final order line, supporting FEFO and Farm/Outlet/Customer traceability.
- Care Plan extensibility: targets use metric_code + value + unit rather than fixed columns, so new clinical metrics do not require repeated schema migrations.
- CRM/Customer 360 is a read model over the shared JK ID and role relationships, not a second master customer table.

3.5 Integration, offline sync & real-time event architecture

Area	Pattern	Control / behavior
Offline POS	Odoo POS native/offline session capability + controlled synchronization	Use Odoo Community POS as the transaction engine. Validate target-version offline behavior, browser/device cache, reconnect and payment constraints during pilot; do not build a second custom POS ledger unless the gap analysis proves it necessary.
Stock conflict	Odoo authoritative stock moves + reconciliation	Odoo remains authoritative for stock/reservation/movement. Custom apps consume stock availability via APIs/read models and never overwrite stock tables. Reconnect conflicts and duplicate submissions use transaction IDs/idempotency plus reconciliation reports.
Cash/shift close	Odoo POS session/payment reconciliation	Outlet cash/session close and payment records reconcile in Odoo. Network recovery, duplicate-order prevention and

Area	Pattern	Control / behavior
		stock/payment reconciliation are explicit pilot acceptance tests.
Wearables / CGM	Health Connect + HealthKit first	Direct vendor API or BLE where required; manual entry always remains available via the same backend endpoint.
Event backbone	RabbitMQ first	Single ingestion path can feed Alert Engine, notifications and Warehouse; Kafka only if proven event volume justifies it.
Notifications	FCM/APNs + WebSocket	Domain events trigger mobile push; WebSocket supports OPD queue and live manager dashboards.
Reliability	Idempotency + retry + trace IDs	Repeat-safe processing, correlation IDs and observable retry behavior for offline and vendor-failure scenarios.
Community feed	Hybrid consistency	Immediate consistency for a user's own write; ranked feed re-scored asynchronously.

Clinical alert flow. Dangerous reading → Alert Engine rule → patient notification and clinician task in parallel → clinician acknowledgement SLA → escalation to on-call supervisor when unacknowledged. The platform provides reliable delivery and an audit trail; it is monitoring infrastructure, not an emergency-response replacement.

3.6 Security, privacy & resilience architecture

Control area	Implementation
Authorization	RBAC scoped by Concern, Location and Function; enforced at gateway/service boundary, not only in UI.
Authentication	Mandatory 2FA for staff/admin; OTP-based consumer/patient login to reduce onboarding friction.
Encryption	TLS in transit, encryption at rest, plus field-level encryption for sensitive EMR/device/Care Plan fields.
Audit	Sensitive health-data access logged on reads as well as writes; correlation/trace IDs retained for investigations.
Consent	Explicit, revocable consent records govern device sharing, community use and analytics/AI processing.
Application security	Dependency scanning, SAST and OWASP controls in CI; independent VAPT before major go-lives and annually thereafter.
Backup / DR	Automated backups, restore tests and module-specific RTO/RPO; DR rehearsal before major public releases.
Local resilience	Secured VPN and optional local edge/cache at Dhaka flagship; other outlets use offline-first POS with background sync.
Observability	Structured logs, trace IDs, metrics, SLO dashboards, vulnerability backlog and monthly cloud/cost review.

3.7 Technology platform & DevOps

Layer	Choice	Reason
Web	Next.js (React) + TypeScript	SSR/SEO, reusable component library, broad hiring pool.
Mobile	Flutter	Single Android/iOS codebase, offline store, BLE/device ecosystem.
Backend	Node.js / NestJS	Strong modular boundaries, TypeScript contracts and API-first service design.
AI / data services	Python / FastAPI	ML/data-science ecosystem and model-serving path.
ERP	Odoo Community (open-source baseline)	Commercial/financial core: Product, Inventory/Warehouse, Procurement, POS, outlet + online/e-commerce sales, customer invoicing, payment/reconciliation and Accounting.
Primary DB	Managed PostgreSQL	ACID for finance/clinical and JSONB for controlled semi-structured targets.
Cache/session	Redis	Session, rate limiting, cache and selected queue/state use cases.
Events	RabbitMQ → Kafka only if justified	Operationally simpler at current scale; future throughput path remains open.
Search	PostgreSQL FTS → OpenSearch when needed	Avoid search-cluster operational cost before catalogue/community volume requires it.
Object/media	S3-compatible object storage + CDN	Food/progress images and education media with compression/lifecycle policies.
Runtime	Managed Kubernetes	Dev/Staging/Prod parity, horizontal scaling and workload isolation.
CI/CD	GitHub Actions + Argo CD (or equivalent)	Traceable commit-to-deployment pipeline with gated production promotion.

3.8 Data & AI architecture

Pipeline step	Architecture	Governance / control
1. Capture	Food photo, user correction, device reading, adherence outcome	Write consent/data-lineage metadata with the event.
2. Recognize	Third-party food-recognition API + locally curated Bangladeshi nutrition data	Buy phase accelerates launch while local data remains JK-owned.
3. Human confirm	Confidence score + correction when below threshold	Manual correction is a first-class product path and safety control.
4. Store & label	Consented events land in warehouse/training dataset	Corrections become labeled examples; preserve source and model/version lineage.
5. Build decision	Local model POC after sufficient labeled data exists	Compare quality, latency and cost before shifting inference in-house.

- Indicative Phase-2 quality target: $\geq 80\%$ correct dish identification with an approximate $\pm 20\%$ macro/portion estimate target; always expose confidence and confirmation when uncertain.
- Inference cost control: cache common/repeated meals and batch non-urgent recognition rather than paying synchronous vendor cost for every photo.
- Warehouse prerequisites: event capture, stable metric_code conventions, consent tagging and data lineage are established before predictive models depend on them.

3.9 Key end-to-end system flows

Flow	Path	Architecture intent
Care Plan	Clinician Console → Clinical Service → PostgreSQL → domain event → App Care Plan API → patient	Clinical remains the authoring source of truth; the App consumes a stable API contract rather than reading clinical tables.
Meal / AI	Super App → Object Storage → recognition API/local nutrition → user confirmation → event → warehouse/adherence → clinician dashboard	Separate media, inference and analytics paths; corrections become labeled data with consent and lineage.
Online / e-commerce sale	Website/Super App → API Gateway / Odoo Adapter → Odoo Sales/e-commerce transaction → payment → Odoo stock/financial posting → domain events/read models	JK owns the customer experience; Odoo Community owns the commercial transaction, inventory impact and financial record.
Outlet / POS sale	Odoo POS → Odoo sales/inventory/accounting → integration events/read models → CRM/BI/support	One POS/commercial source of truth; custom systems consume results rather than duplicating sale or stock ledgers.
Clinical invoice	Custom ERP/Clinical → clinical services/bill approval → integration → Odoo customer invoice/payment/accounting reference	Custom Clinical owns medical/service context; Odoo owns the financial invoice, payment/reconciliation and ledger.
Payroll	Custom ERP Attendance/Payroll → payroll approval → integration → Odoo Finance journal/payable/ledger reference	Custom ERP owns attendance and payroll calculation; Odoo owns the final accounting impact.
Offline POS	Odoo POS offline/session mode → reconnect/sync → Odoo authoritative transaction/stock/payment reconciliation	Offline acceptance is validated against the selected Odoo Community version; idempotency and reconciliation protect recovery.
Authentication	Channel → JK ID/SSO → role/consent resolution → token → API Gateway → custom domain or Odoo adapter	Identity and authorization remain centralized while Odoo customer/employee references are mapped, not duplicated as a competing identity master.

3.10 Deployment topology & non-functional baseline

Area	Baseline design	Control / target
Environment model	Separate Dev, Staging and Production with isolated secrets/configuration	Every deployment traceable to source control; production promotion is gated.
Cloud runtime	Managed Kubernetes + managed PostgreSQL + Redis + RabbitMQ + object storage/CDN	Scale progressively rather than provisioning Phase-5 capacity on day one.
Connectivity	Cloud-first with secured VPN/local resilience for flagship clinical operations	OPD/POS continuity during ISP outage; background synchronization on recovery.
Scale design	11 outlets initially; 500-700 daily patients; 5,000 peak concurrent users; 50,000+ Year-1 registered users	Horizontal scaling and cache/CDN reduce origin pressure as usage grows.
Resilience	Automated backups, restore tests, defined RTO/RPO and DR rehearsal	Clinical/Finance receive stricter recovery targets than community/content workloads.
Observability	Structured logs, metrics, trace IDs, SLO dashboard, alerting	Monthly reliability, vulnerability and cost review with threshold alerts.

3.11 Concise answers to key board questions

These concise answers summarize the controls already defined in the architecture, delivery, risk and operating-cost sections.

Q	Question focus	Concise answer
Q10	Top project-failure risks	Three primary failure drivers to control are: (1) dirty Odoo/Excel/master data - profile early, migrate in stages, reconcile and verify stock before cut-over; (2) clinical workflow churn - embed the Clinical SME, prototype early and sign acceptance criteria before build; (3) capacity/scope pressure - phase hiring, use specialist bridge support when needed, and enforce Definition of Done, change control and quarterly gates.
Q14	Community moderation at scale	Use a hybrid moderation model: technology auto-flags health-misinformation patterns, while humans make moderation decisions through a Moderation Console under an SLA. Moderation tooling and minimum moderation capacity are hard Community go-live gates; automation assists detection, but health-content decisions are not delegated to AI alone.
Q15	Responsibility for dangerous alerts	A dangerous reading triggers the Alert Engine, which notifies the patient and creates the assigned clinician task in parallel. The clinician is accountable within a defined SLA; unacknowledged alerts escalate to the on-call supervisor. Reliability is protected with queued events, idempotent retry, delivery monitoring, trace IDs and an audit trail; the platform is monitoring infrastructure, not the emergency responder.
Q16	Controlling media and AI run cost	Right-size compute with autoscaling limits; compress images,

use CDN caching, adaptive-bitrate video and lifecycle/cold-storage rules; cache repeated/common meal recognition and batch non-urgent AI processing. Track storage/CDN/AI unit costs monthly with budget alerts, and move from Buy to Build only when model quality and economics justify it.

4. 24-Month Delivery Roadmap

Scheduling model. The phases intentionally overlap. Dependencies are respected (identity/master data before integration testing; Clinical authoring before the App Care Plan loop), while independent teams work in parallel to reduce calendar time. UAT happens throughout the program rather than as a final-stage activity.

4.1 Months 1-12

Month	Primary Focus	Major Deliverables	Acceptance / UAT Gate
M1	Mobilize & discover	Steering cadence; architecture decision register; concern-by-concern discovery; access to Odoo/data; baseline NFRs; security/data classification	Approved scope v1; risk register; prioritized backlog
M2	Foundation v1	Cloud accounts; Dev/Staging/Prod; Git/CI/CD; API Gateway skeleton; JK ID model; master-data rules; Odoo Community module/version & gap analysis	Environment readiness review; identity design sign-off; Odoo baseline confirmed
M3	Odoo commercial core starts	Odoo Community Finance/Inventory cleanup; Product/Warehouse/Procurement/POS/Sale s/e-commerce configuration; customer/employee/patient migration tooling; integration contracts; MIS consolidation	Data-quality report; Odoo module gap report; migration rehearsal #1
M4	Odoo POS pilot + Custom ERP/Clinical start	Odoo POS/outlet-sales pilot; stock/lot/FEFO configuration; Website-to-Odoo commerce adapter design; Doctor Profile and Appointment workflow discovery; Website design system	Odoo POS alpha UAT; offline/reconnect tests; clinical workflow sign-off
M5	Commercial + workforce workflows	Odoo POS sale/return/session/payment and online sales order flow; custom ERP ARM, Attendance and Payroll core; patient registration; Appointment/OPD queue; website catalogue/cart APIs	Outlet + online-sales UAT; workforce UAT; clinical desk UAT

Month	Primary Focus	Major Deliverables	Acceptance / UAT Gate
M6	Odoo Commercial Gate 1	Shadow-mode POS pilot; inventory/payment/accounting reconciliation; online order end-to-end test; MIS v1; pilot training; custom ERP workforce baseline	Gate 1: Odoo POS/commercial pilot accepted; reconciliation passed; VAPT #1 / security review
M7	Odoo rollout + Custom ERP/Clinical build	Roll Odoo POS/commercial flow to first outlet wave; EMR core; consent/audit; website frontend connected to Odoo product/stock/sales APIs; unified login/cart	Outlet-wave UAT; online-commerce regression; EMR test cases signed off
M8	App workstream starts	Care Plan authoring; website commerce transaction processing through Odoo; Flutter foundation; notification framework; AI-food API procurement	Website beta; Odoo commerce integration review; mobile architecture review
M9	Clinical pilot + App core	Clinical pilot; website full journey with Odoo sales processing; app login/profile; care-plan read APIs; membership model; clinical billing-to-Odoo posting prototype	Clinical pilot UAT; web/Odoo regression UAT
M10	Clinical + Website v1 go-live	Patient/Doctor/Appointment/OPD/EMR + basic Care Plan live; Group Hub/UOL online commerce processed by Odoo; clinical invoice financial posting operational	Gate 2: production go-live; Odoo/Clinical reconciliation; VAPT #2
M11	Care loop implementation	App planner/tasks; manual food log; reminders; membership/fitness; CRM/Customer 360 read model	App feature UAT; care-team acceptance
M12	AI + device + community foundation	Food recognition buy-phase; Health Connect/HealthKit foundation; community moderation console; analytics events	AI accuracy baseline; moderation go-live checklist

4.2 Months 13-24

Month	Primary Focus	Major Deliverables	Acceptance / UAT Gate
M13	Super App beta	App care plan; nutrition log; membership; CRM; notifications; basic device sync; pilot cohort	Closed beta + clinician/patient UAT
M14	Super App v1 go-live	Care Plan loop end-to-end; Food Log AI; basic wearable/CGM path; operational support model	Gate 3: App v1 accepted; VAPT #3
M15	Community/LMS beta	Community feed + moderation; LMS/course delivery; franchise/distribution workflows; staff-app prototypes	Moderation UAT; franchise process sign-off
M16	Community live + Data platform	Community v1; education v1; central warehouse foundation; event ingestion; BI semantic model	Content-safety gate; data-quality checks
M17	Franchise & staff operations	Franchise/distribution; corporate MOU; Health Tourism/Sinbad integration if API verified; delivery/sample/field staff apps	Integration UAT; field pilot
M18	Phase 4 Gate	Community/Education/Franchise/staff apps release; operating procedures and training	Gate 4: business acceptance; operational handover
M19	Data/BI scale	Warehouse pipelines; advanced role-based dashboards; finance/clinical/commerce KPI layer	BI reconciliation against source systems
M20	Predictive analytics prototype	Adherence/drop-off model prototype; cohort features; alert-quality monitoring; dataset governance	Offline model evaluation; clinical SME review
M21	In-house food model POC	Local dataset training loop; model-serving POC; buy-vs-build cost benchmark	Model quality/cost decision gate
M22	Scale-out	Manager dashboard app; outlet scale support 15-20; performance tuning; autoscaling/budget optimization	Load/performance UAT; DR rehearsal
M23	Hardening & transition	Security hardening; audit cleanup; upgrade rehearsal; ownership transfer from specialists to JK Tech	KT acceptance; runbook audit; resilience test
M24	Program close / steady state	Stabilization; backlog triage; SLO dashboard; annual operating model; Year-3 roadmap	Gate 5: program closure; BAU ownership accepted

5. First 90 Days - Week-by-Week Execution Plan

Week	Focus	Deliverables	Done / Acceptance
W1	Program mobilization	Confirm sponsors/owners, RAID log, stakeholder map, decision cadence, repo ownership, access request list	Kickoff minutes + accountable owner per concern
W2	Current-state & data audit	Odoo modules, Excel sources, website, patient/clinical workflows, master-data duplicates, integration inventory	As-is inventory + migration risk report
W3	Cloud / engineering baseline	Dev/Staging/Prod; network/IAM; secrets; repository; branch/release model; CI skeleton	Environment checklist
W4	Platform skeleton	API Gateway; observability; logging/trace IDs; auth architecture; backup baseline; security standards	Architecture review #1
W5	JK ID / SSO	Party/role model; OTP/staff 2FA; token flow; consent linkage; first end-to-end login	Identity demo + security UAT
W6	Master Data	Product/customer/employee schemas; mapping rules; data cleansing scripts; staged imports	Migration rehearsal #1
W7	Odoo Community stabilization	Finance/inventory gap analysis; product/warehouse/POS/sales/e-commerce module configuration; supported API/integration contract	Odoo remediation + configuration backlog approved
W8	Odoo POS & sales core	POS sale/return/session/payment; product/stock availability; outlet sales and online sales-order API; reconnect/offline test plan	Odoo POS + online sales alpha demo
W9	Offline, idempotency & reconciliation	Validate Odoo POS offline/reconnect behavior; duplicate transaction protection; stock/payment reconciliation; failure/retry rules for Website/App adapter	Network-failure + reconnect test
W10	MIS & migration rehearsal	Sales/stock/accounting MIS; Odoo/Excel reconciliation; pilot outlet master data; custom ERP employee/patient mapping	MIS sign-off + migration rehearsal #2
W11	Pilot outlet + online UAT	Staff training; device/printer/payment tests; outlet POS + online order parallel-run script; support channel	Pilot-readiness checklist
W12	Shadow-mode pilot & Gate 1 prep	Live Odoo POS/commercial shadow usage; defect burn-down; stock/payment/accounting reconciliation; management demo; next-quarter plan	Go/no-go for controlled Odoo commercial cut-over

6. Workstreams, Parallelism & Critical Dependencies

Workstream	Active Window	Scope	Critical Dependency
Foundation / Platform	M1-M24	JK ID, API Gateway, CI/CD, observability, security, notification, integration contracts	Blocks integration testing for every product stream
Odoo Commercial / Financial ERP	M2-M8 build/rollout; then BAU/enhance	Odoo Community Product, Inventory/Warehouse, Procurement, POS, outlet sales, online/e-commerce sales processing, customer invoices, payment/reconciliation, Accounting and MIS	Master data + Odoo module/version validation + stock/payment/accounting reconciliation
Custom ERP / Clinical / Workforce	M4-M10 core; M11+ enhance	Doctor Profile, Patient, Appointment, OPD/EMR, Care Plan, clinical billing workflow, ARM, Attendance and Payroll calculation	Clinical SME + identity + Odoo finance-posting contracts + audit/security
Website	M4-M10	Group hub and customer commerce UX, Health Revolution booking, unified login/cart; product/stock/sales processing through Odoo APIs/adapter	JK ID + Odoo product/stock/sales contracts + booking APIs
Super App / Care Plan	M8-M14	Customer app, care plan loop, food log, membership, devices, notifications	Clinical authoring model must be stable
Community / Education	M12-M18	Feed, moderation, LMS, content/video, expert presence	Moderation console and content policy are hard go-live gates
Franchise / Staff Apps	M14-M18	Distribution, corporate/health tourism integrations, field staff apps	Verified partner APIs + stable inventory/identity
Data / BI / AI	M1 event capture; M16-M24 maturity	Warehouse, dashboards, adherence prediction, local food model evolution	Event/data lineage and consent capture must exist from day one

7. Team Ramp & Responsibility Model

Period	Approx. Team Size	Composition	Primary Delivery Load
M1-2	6.0	Architect, 2 backend, DevOps, 0.5 QA, Odoo specialist, 0.5 UI/UX	Foundation + Odoo discovery
M3-6	8.5	Architect, 3 backend, DevOps, QA, Frontend, Odoo specialist, 0.5 UI/UX	ERP/POS + early Clinical/Website
M7-10	13.5	Architect, 4 backend, DevOps, 2 QA, 2 Frontend, 2 Mobile, Odoo specialist, 0.5 UI/UX	Peak Clinical/Website + App start
M11-14	14.0	Architect, 4 backend, DevOps, 2 QA, 2 Frontend, 3 Mobile, in-house Odoo	Super App / Care Plan peak
M15-18	12.0	Architect, 4 backend, DevOps, 2 QA, Frontend, 2 Mobile, in-house Odoo	Community/Franchise + Data foundation

Period	Approx. Team Size	Composition	Primary Delivery Load
M19-24	11.0	Architect, 3 backend, DevOps, 2 QA, Frontend, 2 Mobile, in-house Odoo	Scale, BI/AI, hardening, KT

Role consolidation. No separate Product Owner, Data/AI Engineer, ML/AI Specialist or Community Moderator headcount is included in this budget; those responsibilities are covered by the Tech Lead / Architect. Clinical/Domain SMEs are seconded from Health Revolution and therefore are not treated as incremental payroll. Their availability remains a hard delivery dependency for workflow and safety sign-off.

8. Cost Model

Costing follows the architecture and delivery model above. The baseline uses the recalculated role assumptions, 15% in-house loading, specialist Odoo/UIUX support, a progressive infrastructure ramp, and a 12% centrally controlled contingency. Product Owner, Data/AI, ML/AI and Community Moderation responsibilities are covered by the Tech Lead / Architect in this baseline.

8.1 Costing basis & numeric assumptions

Why numeric assumptions are required. The baseline document intentionally used relative cost weights instead of fixed currency values. For financial planning, this version converts that shape into a BDT baseline using role-rate assumptions. These rates are placeholders for budgeting and must be replaced by HR offers and signed supplier quotations before procurement.

8.2 Role-rate assumptions

Role	Mode	Planning Rate (BDT/month)	Scope
Tech Lead / Architect	In-house	200,000	Architecture, security, vendor governance, product/backlog ownership, process mapping, acceptance criteria, Data/AI & ML oversight, community moderation governance, final go-live sign-off
Senior Backend Engineer	In-house	100,000	API Gateway, core platform, domain architecture
Mid Backend Engineer	In-house	80,000	Custom ERP/Clinical, ARM/Attendance/Payroll, Odoo integration adapters, membership and community services
Frontend Engineer	In-house	80,000	Next.js website/admin experiences
Flutter Mobile Engineer	In-house	70,000	Super App, offline sync, device integrations
QA / Test Engineer	In-house	50,000	Manual + automation + release/UAT coordination
DevOps / Platform Engineer	In-house	100,000	Cloud, Kubernetes, CI/CD, security baseline, observability
Odoo/ERP Engineer (after transition)	In-house	70,000	Odoo Community Product/Inventory/POS/Sales/e-commerce/Finance extension, supported integrations and maintenance

Role	Mode	Planning Rate (BDT/month)	Scope
Odoo Specialist	Outsourced	70,000 / FTE-month	Initial Odoo Community stabilization and configuration across Finance/Inventory/POS/Sales/e-commerce, integration support and knowledge transfer
UI/UX Designer	Contract	80,000 / month	Design system, low-literacy UX, service flows

8.3 Development and specialist cost by delivery block

Period	In-house Payroll / mo (M BDT)	External / mo (M BDT)	Loaded Dev Cost / mo* (M BDT)	Period Total (M BDT)
M1-2	0.505	0.110	0.691	1.382
M3-6	0.690	0.110	0.904	3.614
M7-10	1.040	0.110	1.306	5.224
M11-14	1.180	0.000	1.357	5.428
M15-18	1.030	0.000	1.185	4.738
M19-24	0.950	0.000	1.093	6.555
TOTAL				26.941

***Loaded development cost. In-house payroll is loaded with 15% for employer overhead/benefits/recruitment friction; external specialist rates are treated as fully loaded. No separate Product Owner, Data/AI Engineer, ML/AI Specialist or Community Moderator FTE is budgeted; those responsibilities are included in the Tech Lead / Architect baseline. The separate recruitment/onboarding budget covers hiring campaigns, joining equipment gaps and exceptional onboarding costs.**

Cost Category	24-Month Base (M BDT)	Planning Note
Loaded engineering + outsourced specialists	26.9	Core technical delivery team plus Odoo/UIUX specialist support; consolidated roles absorbed by Tech Lead
Cloud / platform infrastructure	10.0	Compute, managed DB, Redis/queue, storage/CDN, backup/DR; grows by phase
AI / Nutrition API consumption	1.8	Buy-phase food recognition/nutrition baseline; controlled by caching/batching
Communication / Maps / Monitoring / SaaS	2.4	SMS/email/WhatsApp baseline, maps, APM/logging, low-volume tools
Independent VAPT / security testing	1.8	Three major pre-go-live security assessments at ~0.6M each
Engineering/test equipment & field hardware	2.5	Developer/test devices, mobile devices, POS/peripheral lab and pilot equipment
Data migration, training & change management	2.0	Rehearsals, outlet/clinic training, cut-over support and reconciliation
Recruitment & onboarding reserve	1.0	Hiring campaigns, screening, joining/onboarding costs
Subtotal before contingency	48.4	Approved delivery baseline
Contingency reserve (12%)	5.8	Only released against approved risk/change
TOTAL PROGRAM ENVELOPE	54.3	~BDT 5.43 crore over 24 months

8.4 Infrastructure ramp assumption

Period	Monthly Infra (M BDT)	Period Total (M BDT)	What drives cost
M1-3	0.12	0.36	Foundation/dev/test + small production footprint
M4-6	0.20	0.60	Odoo POS/commercial pilot + early Custom ERP/Clinical/Website
M7-12	0.30	1.80	Odoo commercial production + Website/Clinical + App beta/media
M13-18	0.50	3.00	Super App + food photos + Community/LMS + warehouse
M19-24	0.70	4.20	Higher usage, BI/AI workloads, 15-20 outlet scale support
TOTAL		9.96	Rounded to BDT 10.0M in the program budget

9. Quarterly Budget Release & Delivery Gates

Budget Window	Base Release (M BDT)	Delivery Focus	Release / Acceptance Gate
Q1 - M1-3	4.79	Foundation + identity/master data + Odoo Community baseline start	Architecture baseline, environments live, identity path demonstrated
Q2 - M4-6	5.06	Odoo POS/commercial pilot + Custom ERP/Clinical/Website start	Odoo POS/outlet + online-sales pilot accepted; migration/reconciliation passed; VAPT #1
Q3 - M7-9	6.06	Odoo commercial rollout + Custom ERP/Clinical/Website build + App foundation	Clinical pilot ready; website/Odoo beta flow; App architecture accepted
Q4 - M10-12	7.06	Clinical + Website go-live with Odoo commerce processing; App/Care loop build	Production gate + VAPT #2; Odoo/Clinical reconciliation; web/clinical BAU stable
Q5 - M13-15	6.88	Super App go-live + Community/LMS beta	App v1 accepted + VAPT #3; closed-loop care functioning
Q6 - M16-18	6.33	Community/Education/Franchise + warehouse	Phase-4 business acceptance; operational training complete
Q7 - M19-21	6.11	BI/AI maturity + local model POC	Data reconciliation; model quality/cost decision
Q8 - M22-24	6.15	Scale, DR, hardening, handover	Performance/DR/KT accepted; BAU ownership signed
Program contingency	5.81	Central reserve; not pre-allocated to quarters	Release only through Steering Committee change/risk approval

Budget-control rule. Quarterly funds should be approved against the next delivery gate; monthly burn is reviewed by the Tech Lead and Finance. A forecast variance above 5% requires a written recovery plan; above 10% requires Steering Committee scope/time/budget approval.

10. Quality, UAT, Security & Release Strategy

Cadence / Gate	Required Activities	Exit Condition
Every sprint	Unit/integration tests, API contract checks, targeted business UAT, demo, security/dependency scan	Defects triaged before next sprint planning
Every month	Regression pack, cloud-cost review, vulnerability backlog, data reconciliation checks	Monthly delivery + burn report
Before outlet/clinic rollout	Migration rehearsal, staff training, parallel run, rollback plan, support roster	Business owner cut-over approval
Before major public go-live	Performance/load test, VAPT, DR/backup restore check, observability/SLOs, privacy/consent verification	Formal go/no-go chaired by Tech Lead + Business Owner
Post go-live 2-4 weeks	Hypercare, incident review, adoption metrics, backlog stabilization	Exit hypercare only when SLO/defect thresholds are stable

11. Budget & Timeline Risks with Controls

Risk	Exposure	Impact	Control
Hiring delay	High	Pushes critical path because JK Tech starts with no bench	Start Phase-2/3 hiring during Phase 0-1; keep approved specialist bridge contracts
Dirty Odoo/Excel data	High	Rework, reconciliation failure, rollout delay	Profile data in M1-2; stage migration; physical stock count before outlet cut-over
Clinical workflow churn	High	EMR/Care Plan rework	Embed clinical SME; prototype workflows early; sign acceptance criteria before build
Odoo over-customization	Medium/High	Upgrade/maintenance lock-in or recreating differentiated clinical/HR logic inside Odoo.	Keep Odoo focused on commercial/financial core; keep Clinical/ARM/Attendance/Payroll workflows in custom ERP; use supported APIs/adapters and regression tests for upgrades.
Cloud/media/AI usage spike	Medium	Run-cost exceeds envelope	Compression/lifecycle, caching/batching, quotas, monthly unit-cost dashboard
Vendor/API availability	Medium	Blocks Sinbad, wearable or AI features	Adapter interfaces; manual fallback; defer integrations that are not verified
Scope expansion	High	Budget and timeline drift	Definition of Done + change control; contingency is not a scope-slush fund
Security/compliance defect	High	Go-live delay and business risk	Security by design, field encryption, read-audit for health data, independent VAPT gates

12. Limited-Budget / Half-Year-1-Budget Fallback

- Defer Community Platform and Education/LMS before cutting the shared foundation, Odoo commercial core, or Custom ERP/Clinical.
- Delay deep Wearable/CGM integration; keep manual entry and basic Health Connect/HealthKit capability where affordable.
- Keep Food Recognition in buy-mode longer and delay the in-house model build; this preserves user value while avoiding early model investment.
- Defer Franchise Management and Health Tourism/Sinbad integration until the core operating system is stable.

- Do not cut JK ID/SSO, API Gateway, Master Data, core security (2FA/encryption/audit), Odoo commercial/financial core, or Custom ERP/Clinical foundation.

13. Program Governance & Reporting

Cadence	Meeting / Control	Participants	Output
Daily	Engineering stand-up per workstream	Engineering leads	Blockers, defects, environment issues
Weekly	Architecture/vendor/risk review	Tech Lead + DevOps + module leads	RAID, ADRs, dependency plan, burn vs plan
Bi-weekly	Sprint review + UAT + planning	Business owners + engineering	Accepted increments and next sprint commitment
Monthly	Steering committee	GMO/CEO/CFO/Tech Lead	Budget, timeline, scope changes, risk decisions
Quarterly	Phase gate / budget release	Steering committee	Acceptance gate, next-quarter budget, hiring/vendor decisions

14. Architect Recommendation

Adopt the architecture-led 24-month roadmap with quarterly funding gates and a BDT 54.3M planning ceiling, while treating BDT 48.4M as the delivery baseline and BDT 5.8M as controlled contingency. Preserve the synchronized technical model: one JK ID, Odoo Community as the commercial/financial system of record, a custom JK ERP/Clinical platform for Doctor/Patient/Appointment/EMR/Care Plan/Clinical Billing/ARM/Attendance/Payroll workflow, API/event integration between them, security/consent controls by design and gated releases. Stabilize master data and Odoo commercial operations, establish Custom ERP/Clinical, then scale App, Community and AI capabilities on top of trusted operational data. The costing assumptions remain unchanged pending the M1-M2 Odoo module/gap validation.

- Approve Phase 0 and Q1 budget immediately only after named business owners and data-access owners are confirmed.
- Validate all remaining role rates with HR and convert Odoo/UIUX placeholders into signed quotations before Q2 approval; validate any Odoo hosting/add-on/support and AI/API vendor consumption separately as operating/vendor costs.
- Lock Odoo Community as the baseline and validate the exact supported version/modules, hosting model, POS offline/reconnect behavior, e-commerce/Sales APIs, backup/DR approach and any paid add-on/support requirements before commercial production scaling.
- Use business acceptance and measurable technical gates (reconciliation, VAPT, performance, DR, UAT) as the condition for go-live and next-quarter budget release.
- Review the cost model every quarter using actual salary, cloud unit cost, AI cost per food-log, SMS cost per active patient and storage/CDN cost per active user.

Appendix A - Budget Formula & Reforecast Method

Cost Driver	Formula	Control
People cost	Sum(role count x monthly rate x active months) + 15% in-house loading	Reforecast on hiring date/rate changes; no separate PO/Data-AI/ML/moderator FTE in baseline
Cloud cost	Compute + DB + cache/queue + object storage + CDN + backup/DR	Track actual monthly unit cost; forecast from active users/content volume
AI cost	Recognition API calls x unit price + nutrition data + model serving compute	Track cost per confirmed food-log and cache hit rate
Communication	SMS/WhatsApp/email message volume x unit rate	Track messages per active customer/patient
Contingency	12% of approved baseline	Not part of team discretionary spend; requires change/risk approval

Final validation checklist. Before approval, Finance/HR should validate salary bands; Procurement should obtain Odoo Community implementation/hosting/add-on/support, AI/SMS/cloud/VAPT quotations; Business Owners should confirm scope and migration volumes; the Odoo pilot must validate POS offline/reconnect, outlet + online sales, inventory, payment/reconciliation and accounting flows; Legal/Compliance should confirm health-data residency and consent requirements. Material variances roll into the next quarterly budget gate.